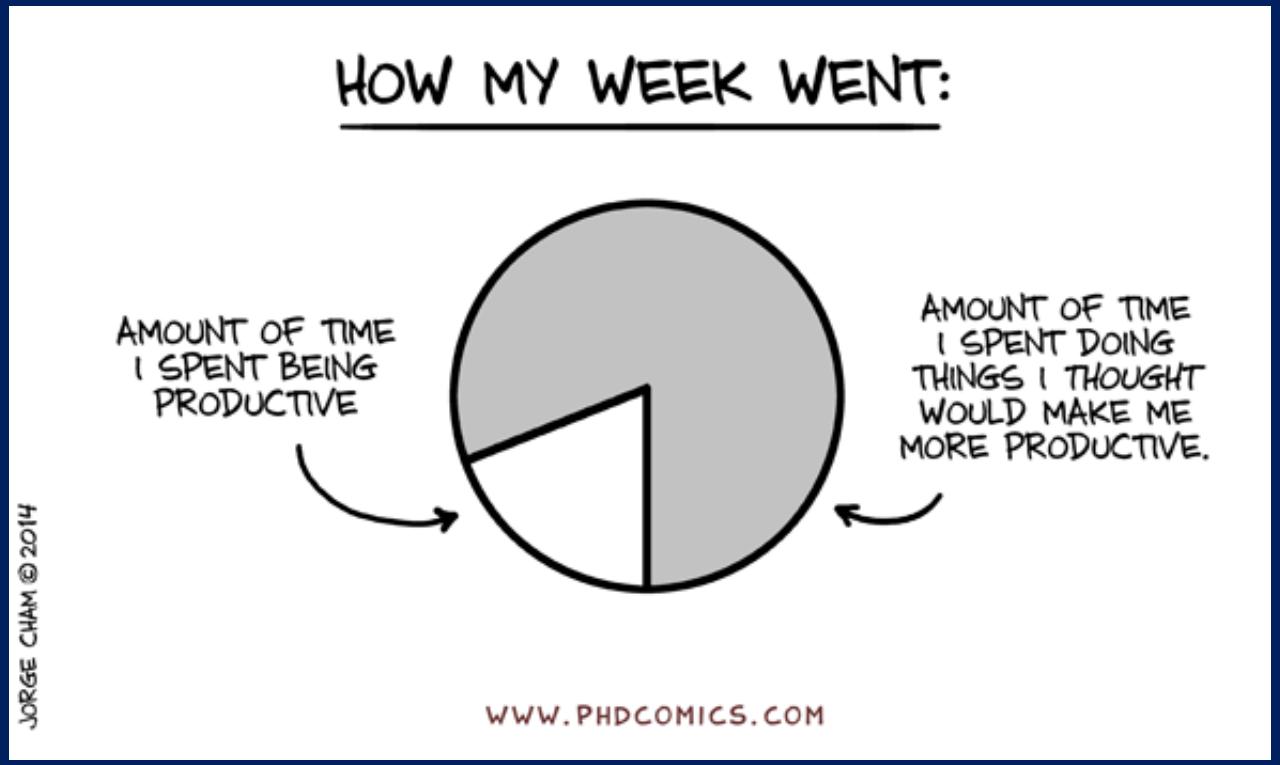
Research Skills Class 1:

Project planning



What is “project planning”?

- Having a **structured approach** to planning and handling tasks, using simple tools and habits, so you can deliver what’s needed and make your life easier!



Project planning “survival guide”!

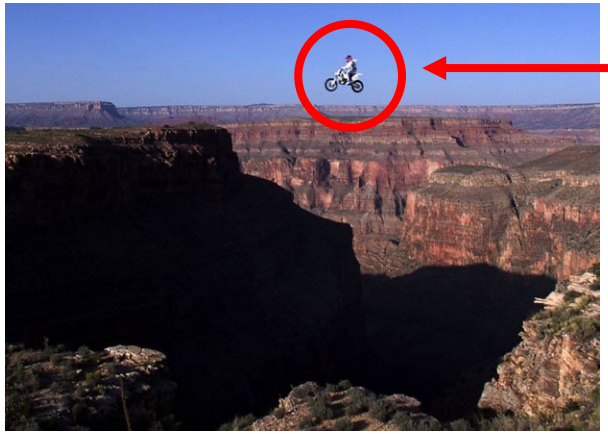
What do you need …??



- Split project into chunks/milestones
- Priorities for each
- Project timeline
- Method for recording progress

*If your project has “risks” involved, e.g., if you need to access to lasers or other equipment which is not guaranteed, you should probably also think about “contingency planning”

Breaking it down



Many of us approach projects like this ☺

A better approach is to break down your big goal into smaller chunks

- Makes the goal feel more manageable
- Helps track your progress and avoid last-minute scrambles
- Easier to get feedback on each part (helps your supervisor help you)



You'll need to know when each chunk is finished – using "milestones"

Let's see an example …

Project goal: *"Measure the growth rate of cosmic structure using galaxy motions"*

??

Breaking it down into chunks:

- Review the literature which explains why my work is interesting
- Download galaxy dataset and understand the meaning of the columns
- Create a computer code to measure correlation function of the dataset
- Research the theoretical equations to compare with the measurement
- Create a computer code to evaluate equations and fit the growth rate
- Write your thesis!

Let's see an example …

Project goal: *"Measure the growth rate of cosmic structure using galaxy motions"*

??

Milestones for each chunk:

I have listed the key contributions of my work to the field

I can plot different columns of the data and the results make sense

I have successfully validated the code using a test dataset

I have written the equations in a document

I have fit for the parameter and validated the result

Thesis is submitted ☺

Let's see an example …

Project goal: *"Measure the growth rate of cosmic structure using galaxy motions"*

??

Priorities for each chunk:

Review the literat… which explains why my work is interesting — 4

Download galaxy da… and understand the meaning of the columns — 2

Create a computer … to measure correlation function of the dataset — 3

Research the theore… equations to compare with the measurement — 5

Create a computer … to evaluate equatio… and fit the growth rate — 6

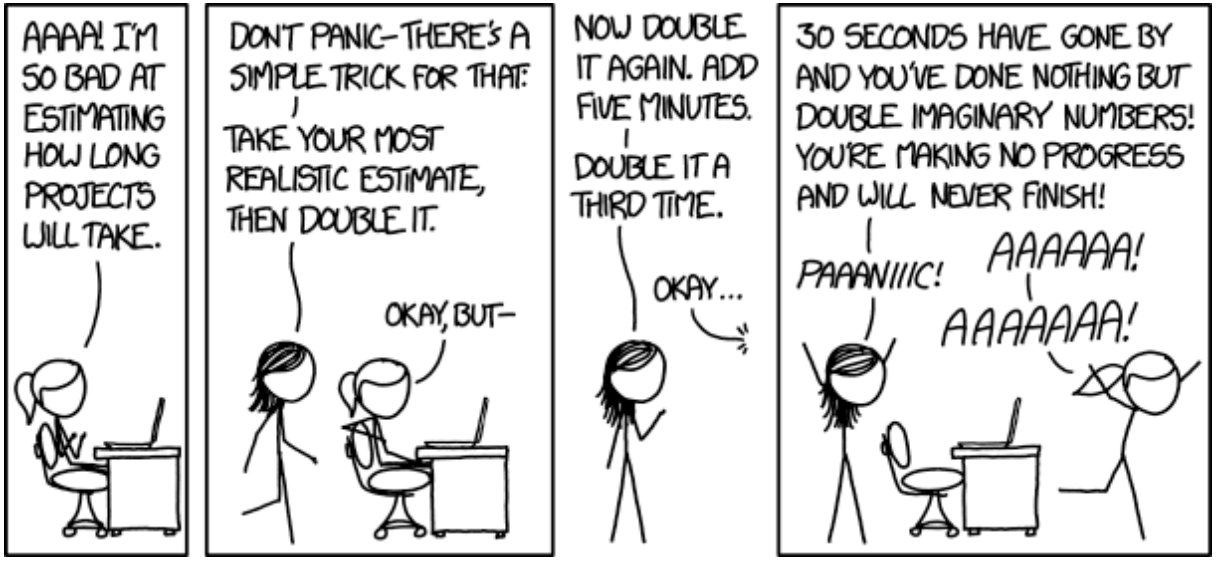
Write your thesis! — 1

The power of “no” !



- **Prioritisation** can be very helpful in judging when to include or exclude a potential new research task
- If you add an extra task, it means not doing something else – **how do the priorities compare?**

The dreaded timeline



- Now we've identified the parts of our project, we need to plan **how much time** to spend on each
- Common time underestimates are **coding** and **writing**
- **Simple visualization of a timeline** is often effective
  (This is sometimes called a "Gantt chart" ☺)

The dreaded timeline

Here is a simple example based on the above project:

Weeks of semester →

Activity	Priority	1	2	3	4	5	6	7	8	9	10	11	12
Literature review	4												
Understand dataset	2												
Measurement code	3												
Understand theory	5												
Fit theory	6												
Write thesis	1												

The dreaded timeline

Here is a simple example based on the above project:

Weeks of semester →

Activity	Priority	1	2	3	4	5	6	7	8	9	10	11	12
Literature review	4												
Understand dataset	2												
Measurement code	3												
Understand theory	5												
Fit theory	6												
Write thesis	1												

Can turn these green as you achieve them!

Which tasks can happen simultaneously?

Which tasks need to be finished before others can start?

What happens when it goes wrong?

- Research and life are unpredictable, and timelines often don't work out – **this is normal**!
- You'll need to **regularly re-visit and refine your goals and timelines** ☺
- It's better to do this in a planned way as the semester unfolds, rather than under pressure at the end!
- Easy to do this using the **priorities** you recorded
- It's best to **protect your writing time** as much as possible – usually better to change the project aims

What happens when it goes ~~wrong~~ normally?

Old goal: *"Measure the growth rate of cosmic structure using galaxy motions"*

Activity	Priority	1	2	3	4	5	6	7	8	9	10	11	12
Literature review	4												
Understand dataset	2												
Measurement code	3												
Understand theory	5												
Fit theory	6												
Write thesis	1												

Ooops, this high-priority task took 7 weeks rather than 4 weeks. Installing python a nightmare. Data is corrupted. Computer is broken. First attempt at algorithm would take 100 years to run.

What happens when it goes ~~wrong~~ normally?

New goal: *"Measure the ~~growth rate of cosmic structure~~ correlations between galaxy motions"*

Activity	Priority	1	2	3	4	5	6	7	8	9	10	11	12
Literature review	4												
Understand dataset	2												
Measurement code	3												
Understand theory	5												
~~Fit theory~~	~~6~~												
Write thesis	1												

This activity can be removed because it's the lowest priority

This activity is unchanged because it's the highest priority

Document and communicate!

- Maintain a **log-book** (paper or digital!) where you track your progress, decisions and intermediate results
- Meet at least **once a week** with your supervisor
- **Write up as you go!** (whilst fresh in your mind)



Staying focused and motivated

- The typical state of research is for something not to be working ☺. **How can you stay motivated?**
- Break tasks into smaller chunks which are more manageable. *Enjoy the "small wins"!*
- Work for a focused period and then take a break or enjoy a reward (e.g. search for "Pomodoro technique")
- Identify the style of research activity you find most enjoyable or rewarding
- Enjoy the community with your fellow students and research group. *Everyone feels the same way!*

Activities

- Create a list of chunks, milestones and priorities for your project
- Arrange these pieces onto a visual timeline
- If your project has any “risky” parts (e.g. access to specialized equipment), write down some potential alternatives

That’s all for today!

Example for an experimental project

Project goal: *"Synthesise spherical crystalline silicon nanoparticles of size 100 – 200 nm"*

??

Breaking it down into chunks:

- Review the literature which explains why my work is interesting
- Learn the laser ablation nanoparticle synthesis experimental apparatus
- Create and conduct experiments to test size & sphericity of particles
- Research theoretical equations to optically characterize the particles
- Create a computer code to simulate the expected results
- Write your thesis!

Example for an experimental project

Project goal: *"Synthesise spherical crystalline silicon nanoparticles of size 100 – 200 nm"*

??

Milestones for each chunk:

I have listed the key contributions of my work to the field

I can operate laser ablation setup to create nanoparticles

I have set up apparatus to test the sphericity and size of the particles

I've simulated all the theoretical equations to interpret the results

I have conducted experiments and validated with theory

Thesis is submitted ☺

Example for an experimental project

Project goal: *"Synthesise spherical crystalline silicon nanoparticles of size 100 – 200 nm"*

??

Weeks of semester →

Activity	Priority	1	2	3	4	5	6	7	8	9	10	11	12
Literature review	4												
Learn the set-up	2												
Create new set-up	3												
Conduct experiment	5												
Fit theory	6												
Write thesis	1												